

**SAFETY DATA SHEET**

Creation Date 22-September-2009

Revision Date 24-December-2021

Revision Number 5

**1. Identification**

**Product Name** Adipic dihydrazide

**Cat No. :** AC203310000; AC203310250; AC203311000; AC203315000

**CAS-No** 1071-93-8  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company**

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Manufacturer**  
Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

**2. Hazard(s) identification****Classification**

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |            |
|---------------------------------------------------------|------------|
| <b>Skin Sensitization</b>                               | Category 1 |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3 |
| Target Organs - Respiratory system.                     |            |

**Label Elements**

**Signal Word**  
Warning

**Hazard Statements**  
May cause an allergic skin reaction  
May cause respiratory irritation

**Precautionary Statements****Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray

Use only outdoors or in a well-ventilated area

Contaminated work clothing should not be allowed out of the workplace

**Response**

IF ON SKIN: Wash with plenty of soap and water

IF INHALED: Remove person to fresh air and keep comfortable for breathing

Call a POISON CENTER/ doctor if you feel unwell

If skin irritation or rash occurs: Get medical advice/attention

Wash contaminated clothing before reuse

**Storage**

Store in a well-ventilated place. Keep container tightly closed

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Toxic to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component                     | CAS-No    | Weight % |
|-------------------------------|-----------|----------|
| Hexanedioic acid, dihydrazide | 1071-93-8 | >95      |

### 4. First-aid measures

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms/effects**

None reasonably foreseeable. . May cause allergic skin reaction. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

**Notes to Physician**

Treat symptomatically

### 5. Fire-fighting measures

**Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Unsuitable Extinguishing Media** No information available

**Flash Point** No information available  
**Method -** No information available

**Autoignition Temperature** No information available  
**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

#### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

#### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>).

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### NFPA

**Health**  
2

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
N/A

### 6. Accidental release measures

#### Personal Precautions

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

#### Environmental Precautions

Do not flush into surface water or sanitary sewer system.

#### Methods for Containment and Clean Up

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

### 7. Handling and storage

#### Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation.

#### Storage.

Keep container tightly closed in a dry and well-ventilated place. Store in freezer. Incompatible Materials. Strong oxidizing agents. Strong bases. Strong acids.

### 8. Exposure controls / personal protection

#### Exposure Guidelines

This product does not contain any hazardous materials with occupational exposure limit established by the region specific regulatory bodies.

#### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Personal protective equipment

##### Eye Protection

Wear safety glasses with side shields (or goggles) Goggles

##### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers | -               | Splash protection only |
| Neoprene       | recommendations   |                 |                        |
| Natural rubber |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

Do not allow material to contaminate ground water system. Prevent product from entering drains.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                                 |
|----------------------------------------|---------------------------------|
| Physical State                         | Solid                           |
| Appearance                             | White, Light yellow             |
| Odor                                   | Odorless                        |
| Odor Threshold                         | No information available        |
| pH                                     | 6.5-8.0 1% aq.sol               |
| Melting Point/Range                    | 178 - 182 °C / 352.4 - 359.6 °F |
| Boiling Point/Range                    | No information available        |
| Flash Point                            | No information available        |
| Evaporation Rate                       | Not applicable                  |
| Flammability (solid,gas)               | No information available        |
| Flammability or explosive limits       |                                 |
| Upper                                  | No data available               |
| Lower                                  | No data available               |
| Vapor Pressure                         | No information available        |
| Vapor Density                          | Not applicable                  |
| Specific Gravity                       | No information available        |
| Solubility                             | Soluble in water                |
| Partition coefficient; n-octanol/water | No data available               |
| Autoignition Temperature               | No information available        |
| Decomposition Temperature              | No information available        |
| Viscosity                              | Not applicable                  |
| Molecular Formula                      | C6 H14 N4 O2                    |
| Molecular Weight                       | 174.2                           |

## 10. Stability and reactivity

|                 |                                              |
|-----------------|----------------------------------------------|
| Reactive Hazard | None known, based on information available   |
| Stability       | Stable under recommended storage conditions. |

**Conditions to Avoid** Incompatible products. Excess heat. Avoid dust formation.

**Incompatible Materials** Strong oxidizing agents, Strong bases, Strong acids

**Hazardous Decomposition Products** Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Nitrogen oxides (NO<sub>x</sub>)

**Hazardous Polymerization** Hazardous polymerization does not occur.

**Hazardous Reactions** None under normal processing.

## 11. Toxicological information

### Acute Toxicity

**Product Information** No acute toxicity information is available for this product

### Component Information

| Component                     | LD50 Oral     | LD50 Dermal      | LC50 Inhalation             |
|-------------------------------|---------------|------------------|-----------------------------|
| Hexanedioic acid, dihydrazide | >5 g/kg (Rat) | >2 g/kg (rabbit) | LC50 > 5.3 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** May cause sensitization by skin contact

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                     | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------------------|-----------|------------|------------|------------|------------|------------|
| Hexanedioic acid, dihydrazide | 1071-93-8 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component                     | Freshwater Algae | Freshwater Fish                                | Microtox   | Water Flea |
|-------------------------------|------------------|------------------------------------------------|------------|------------|
| Hexanedioic acid, dihydrazide | Not listed       | LC50: > 100 mg/L, 96h static (Cyprinus carpio) | Not listed | Not listed |

|                                      |                                                                          |
|--------------------------------------|--------------------------------------------------------------------------|
| <b>Persistence and Degradability</b> | Soluble in water Persistence is unlikely based on information available. |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                                |
| <b>Mobility</b>                      | . Will likely be mobile in the environment due to its water solubility.  |

| Component                     | log Pow |
|-------------------------------|---------|
| Hexanedioic acid, dihydrazide | -2.41   |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| UN-No                | UN3077                                              |
| Proper Shipping Name | Environmentally hazardous substances, solid, n.o.s. |
| Technical Name       | Hexanedioic acid, dihydrazide                       |
| Hazard Class         | 9                                                   |
| Packing Group        | III                                                 |

#### TDG

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| UN-No                | UN3077                                              |
| Proper Shipping Name | Environmentally hazardous substances, solid, n.o.s. |
| Hazard Class         | 9                                                   |
| Packing Group        | III                                                 |

#### IATA

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| UN-No                | UN3077                                              |
| Proper Shipping Name | Environmentally hazardous substances, solid, n.o.s. |
| Hazard Class         | 9                                                   |
| Packing Group        | III                                                 |

#### IMDG/IMO

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| UN-No                | UN3077                                              |
| Proper Shipping Name | Environmentally hazardous substances, solid, n.o.s. |
| Hazard Class         | 9                                                   |
| Packing Group        | III                                                 |

### 15. Regulatory information

#### International Inventories

| Component                     | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-------------------------------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Hexanedioic acid, dihydrazide | 1071-93-8 | X   | -    | X    | ACTIVE                                        | 213-999-5 | -      | -   |

| Component                     | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-------------------------------|-----------|-------|----------|------|------|------|------|-------|-------|
| Hexanedioic acid, dihydrazide | 1071-93-8 | X     | KE-18695 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations****Authorisation/Restrictions according to EU REACH****Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                     | CAS-No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------------------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Hexanedioic acid, dihydrazide | 1071-93-8 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component                     | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Hexanedioic acid, dihydrazide | 1071-93-8 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information****Prepared By**

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24-December-2021

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24-December-2021

**Revision Summary**

This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**